www.Padasalai.Net www.TrbTnpsc.com

Class : 12

Register Number

COMMON HALF YEARLY EXAMINATION, 2025-26

MATHEMATICS

Time Allowed : 3.00 Hours] [Max. Marks : 90

PART - I

I. Answer all the questions: 20x1=20

1. If A, B and C are invertible matrices of some order, then which one of the following is not true?
   a) $\text{adj}\, A = |A|A^{-1}$ b) $\text{adj}\,(AB) = (\text{adj}\, A)(\text{adj}\, B)$
   c) $\det A^{-1} = (\det A)^{-1}$ d) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
2. Which of the following is / are false, in the case of a plane passing through three points whose position vectors are $\vec{a}$, $\vec{b}$ and $\vec{c}$?
   (i) $[\vec{r} - \vec{a}\ \vec{b} - \vec{a}\ \vec{c} - \vec{a}] = 0$ (ii) $[\vec{r} - \vec{a}\ \vec{a} - \vec{b}\ \vec{c} - \vec{a}] = 0$
   (iii) $[\vec{r} - \vec{a}\ \vec{b} - \vec{a}\ \vec{a} - \vec{c}] = 0$ (iv) $[\vec{r} - \vec{a}\ \vec{a} - \vec{b}\ \vec{a} - \vec{c}] = 0$
   a) (ii) and (iii) b) (iii) and (iv) c) all d) none
3. The radius of the circle passing through the point (6, 2) two of whose diameters are x+y=6 and x + 2y = 4 is
   a) 10 b) $2\sqrt{5}$ c) 6 d) 4
4. If $\frac{\Gamma(n+2)}{\Gamma r(n)} = 90$ then n is
   a) 10 b) 5 c) 8 d) 9
5. In the case of Cramer's rule which of the following are correct?
   i) $\Delta = 0$ ii) $\Delta \neq 0$ iii) the system has unique solution
   iv) the system has infinitely many solutions
   a) (i) and (iv) b) (ii) and (iii) c) all d) none
6. If $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$, where $\vec{a}$, $\vec{b}$, $\vec{c}$ are any three vectors such that $\vec{b} \cdot \vec{c} \neq 0$ and $\vec{a} \cdot \vec{b} \neq 0$, then $\vec{a}$ and $\vec{c}$ are
   a) perpendicular b) parallel
   c) inclined at an angle $\pi/3$ d) inclined at an angle $\pi/6$
7. The order of the differential equation of all circles with centre at (h , k) and radius 'a' is
   a) 2 b) 3 c) 4 d) 1
8. The distance between the planes x + 2y + 3z + 7 = 0 and 2x + 4y + 6z + 7 = 0 is
   a) $\frac{\sqrt{7}}{2\sqrt{2}}$ b) $\frac{7}{2}$ c) $\frac{\sqrt{7}}{2}$ d) $\frac{7}{2\sqrt{2}}$
9. The principal argument of $\frac{3}{-1+i}$ is
   a) $\frac{-5\pi}{6}$ b) $\frac{-2\pi}{3}$ c) $\frac{-3\pi}{4}$ d) $\frac{-\pi}{2}$
10. A random variable X has binomial distribution with n = 25 and p = 0.8 then standard deviation of X is
    a) 6 b) 4 c) 3 d) 2

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

11. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ having centre at (0 , 3) is

a) $x^2 + y^2 - 6y - 7 = 0$ b) $x^2 + y^2 - 6y + 7 = 0$

c) $x^2 + y^2 - 6y - 5 = 0$ d) $x^2 + y^2 - 6y + 5 = 0$

12. The value of $\int_{-1}^{2} |x|\, dx$

a) $\frac{1}{2}$ b) $\frac{3}{2}$ c) $\frac{5}{2}$ d) $\frac{7}{2}$

13. If sin x is the integrating factor of the linear differential equation $\frac{dy}{dx} + Py = Q$, then P is

a) log sin x b) cos x c) tan x d) cot x

14. Identify the incorrect statement

a) $|z|^2 = 1 \Rightarrow {}^1/_z = \overline{z}$ b) $Re(z) \leq |z|$

c) $||z_1| - |z_2|| \geq |z_1 + z_2|$ d) $|z^n| = |z|^n$

15. The number of real numbers in [0 , 2π] satisfying $\sin^4 x - 2\sin^2 x + 1$ is

a) 2 b) 4 c) 1 d) ∞

16. If $v(x , y) = \log (e^x + e^y)$, then $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y}$ is equal to

a) $e^x + e^y$ b) $\frac{1}{e^x + e^y}$ c) 2 d) 1

17. $\cos^{-1}(\cos x) = x$ if

a) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ b) $0 < x \leq \pi$ c) $0 \leq x \leq \pi$ d) $-1 \leq x \leq 1$

18. The tangent to the curve $y^2 - xy + 9 = 0$ is vertical when

a) $y = 0$ b) $y = \pm\sqrt{3}$ c) $y = {}^1/_2$ d) $y = +3$

19. In the set Q define $a \odot b = a + b + ab$. For what value of y, $3 \odot (y \odot 5) = 7$?

a) $y = \frac{2}{3}$ b) $y = \frac{-2}{3}$ c) $y = \frac{-3}{2}$ d) $y = 4$

20. One of the closest points on the curve $x^2 - y^2 = 4$ to the point (6, 0) is

a) (2 , 0) b) $(\sqrt{5}, 1)$ c(3, $\sqrt{5}$) d) $(\sqrt{13}, -\sqrt{3})$

PART - II

II. **Answer any 7 Questions. Question Nunmber 30 is compulsory.** **7x2=14**

21. If $x^2 + 2(k + 2)x + 9k = 0$ has equal roots, Find K.

22. Obtain the equation of the circle for which (3 , 4) and (2 , -7) are the ends of a diameter.

23. Find the differential equation of the curve represented by $xy = ae^x + be^{-x} + x^2$.

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

24. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal

25. Evaluate $\int_0^{\pi/2} \cos^7 x \, dx$

26. Find the intercepts cut off by the plane $\vec{r} \cdot (6\hat{i} + 4\hat{j} - 3\hat{k}) = 12$ on the coordinate axes.

27. The temperature in celsius in a long rod of length 10m, insulated at both ends, is a function of length x given by $T = x(10-x)$. Prove that the rate of change of temperature at the midpoint of the rod is zero.

28. A pair of fair dice is rolled once. Find the probability mass function to get the number of fours.

29. Let $g(x) = x^2 + \sin x$. Calculate the differential dg.

30. Prove that $\sum_{i=1}^{204} (i^{n+1} - i^{n+2}) = 0$

PART - III

III. Answer any 7 Questions. Question Number 40 is compulsory. 7x3=21

31. Show that the polynomial $9x^9 + 2x^5 - x^4 - 7x^2 + 2 = 0$ has at least six imaginary roots.

32. If $|z| = 2$, show that $8 \leq |z + 6 + 8i| \leq 12$.

33. Prove that $[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] = [\vec{a}, \vec{b}, \vec{c}]^2$.

34. Find the equation of the circle whose centre is (3 , 2) and radius is the radius of $x^2 + y^2 + 2x + 4y - 4 = 0$.

35. Two balls are chosen randomly from an urn containing 8 white and 4 black balls. Suppose that we win Rs.30 for each black ball selected and the lose Rs.20 for each white ball selected. If X denotes the winning amount, then find the values of X and number of points in its inverse images.

36. If $A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.

37. Show that the percentage error in the nth root of a number is approximately $1/n$ times the percentage error in the number.

38. Prove that $\tan^{-1}\frac{2}{11} + \tan^{-1}\frac{7}{24} = \tan^{-1}\frac{1}{2}$

39. Verify whether the following compound propositions are tautologies or contradictions or contingency. $((p \vee q) \wedge \neg p) \rightarrow q$

40. Solve the following differential equation : $\frac{dy}{dx} = \sqrt{\frac{1 - y^2}{1 - x^2}}$

PART - IV

IV. Answer all the questions 7x5=35

41. a) Find the value of k for which the equtions $kx - 2y + z = 1$, $x - 2ky + z = -2$, $x - 2y + kz = 1$ have (i) no solution (ii) unique solution (iii) infinitely many solution. **(OR)**

b) Prove by vector method that $\sin(\alpha + \beta) = \sin\alpha \cos\beta + \cos\alpha \sin\beta$.

Kindly Send Me Your Study Materials To Us.Email ID: padasalai.net@gmail.com

42. a) If $z = x + iy$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, show that $x^2 + y^2 = 1$ **(OR)**

b) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following: $\frac{(x+1)^2}{100} + \frac{(y-2)^2}{64} = 1$

43. a) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $1/3$ is a solution.

(OR)

b) Salt is poured from a conveyer belt at a rate of 30 cubic metre per minute forming a conical pile with a circular base whose height and diameter of base are always equal. How fast is the height of the pile increasing when the pile is 10 metre high?

44. a) Find the number of solutions of the equation $\tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1} 3x$.

(OR)

b) Find the parametric form of vector and Cartesian equations of the plane containing the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + t(2\hat{i} - \hat{j} + 4\hat{k})$ and perpendicular to plane. $\vec{r}.(\hat{i} + 2\hat{j} + \hat{k}) = 8$.

45. a) $u = \sec^{-1}\left(\frac{x^3 - y^3}{x + y}\right)$ then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\cot u$.

(OR)

b) In a murder investigation, a corpse was found by a detective at exactly at exactly 8 P.M. Being alert, the detective also measured the body temperature and found it to be 70°F. Two hours later, the detective measured the body temperature again and found it to be 60°F. If the room temperature is 50°F and assuming that the body temperature of the person before death was 98.6°F, at what time did the murder occur? [log (2.43) = 0.88789; log (0.5) = -0.69315]

46. a) A rectangular page is to contain 24cm² of print. The margins at the top and bottom of the page are 1.5 cm and the margins at other sides of the page is 1 cm. What should be the dimensions of the page so that the area of the paper used is minimum.

(OR)

b) Let A be Q - {1}. Define * on A by $x * y = x + y - xy$. Is * a binary on A. If so, examine the closure, commutative, associative, the existence of identity and existence of inverse properties.

47. a) Find the area of the region bounded by $y = \cos x$, $y = \sin x$, the lines $x = \pi/4$ and $x = 5\pi/4$.

(OR)

b) Find the mean and variance of a random variable X, Whose probability density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com